1.4 Modeling with Linear Functions

Name _____ Period ____ Date _____

Algebra II 1.4 · Student sheet

I can…

- write equations of linear functions from graphs and tables.
- find a line of fit and the line of best fit.
- interpret the correlation coefficient and use linear models.

Part 1 · Study time

Hours studied	1	2	3	4	5	6	7	8
Score	62	68	71	75	80	83	88	90

Plot the data, draw a line of fit, and choose two points ON your line: (_____) and (_____)

Your equation: _____ Technology: y = _____ r = _____

What does the slope mean? _____

Part 2 · Notes: fill in the blanks

Slope m = _____; y = mx + b; $y - y_1$ = _____

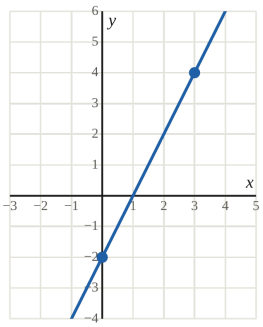
A line of fit is drawn close to _____ of the points.

r near 1: _____. r near −1: _____. r near 0: _____.

Part 3 · Examples (work them with the class)

Example 1

Write an equation of the line.



Example 2

The table shows the total cost of a gym membership. Write an equation for the cost y after x months. What do the slope and y-intercept mean?

Months, x	0	2	4	6
Cost, y ($)	40	100	160	220

Example 3

Use technology to find the line of best fit for the study-time data and the correlation coefficient. Predict the score for 10 hours of studying.

Hours studied	1	2	3	4	5	6	7	8
Score	62	68	71	75	80	83	88	90

Example 4

Describe the correlation shown by each correlation coefficient.

a. $r = -0.94$ b. $r = 0.12$ c. $r = 0.85$

Part 4 · Partner practice

Partner 1

Write an equation of the line through $(-1, 7)$ and $(3, -1)$.

Partner 2

Write an equation of the line through (6, 1) that is perpendicular to $y = 3x - 2$.